

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5101

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I N.AKSHAYA (192425129) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: N.Akshaya

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Annexure A

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

- a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- a. Deduce the most likely original message.
- b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

- a. Demonstrate how the encryption process transforms the message step-by-step.
- b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- a. Analyze how investigators could detect the presence of hidden data within the image.
- b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25 <i>25</i>	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30 <i>25</i>	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20 <i>20</i>	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15 <i>15</i>	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10 <i>10</i>	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

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1) Shift Cipher - Event Communication Security

Definition:

A Shift Cipher is a classical encryption technique in which each letter of the plaintext is shifted by a fixed number of positions in the alphabet.

Formula:

* Encryption: $C = (P + K) \bmod 26$

* Decryption: $P = (C - K) \bmod 26$

Given

Message: MEET AT NOON

Shift = 4 positions

a) Reconstruct the encrypted

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

Encryption: $C = (P + K) \bmod 26$

plaintext	+4	Ciphertext
M	+4	Q
E	+4	I
E	+4	I
T	+4	X
A	+4	E
T	+4	X
N	+4	R
O	+4	S
O	+4	S
N	+4	R

Encrypted message : QIIX EX RSSR

MEET AT NOON
↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓
QIIX EX RSSR

by Retrieve the original message

* The authorized member knows the shift is 4.

* Shift every ciphertext letter 4 positions backward

Decryption: $p = (c - k) \bmod 26$

Ciphertext - 4 = plaintext

Ciphertext	-4	plaintext
Q	-4	M
I	-4	E
I	-4	E
X	-4	T
E	-4	A
X	-4	T
R	-4	M
S	-4	O
S	-4	O
R	-4	N

QIIX EX RSSR → MEET AT NOON

Final Answer:

Encryption: MEET AT NOON → QIIX EX RSSR

Decryption: QIIX EX RSSR → MEET AT NOON

3) Secure Messaging System - Vignere Cipher

Definition

The Vignere Cipher is a polyalphabetic Substitution Cipher that uses a keyword to encrypt a message.

Given

* plaintext: HELLO

* keyword: KEY

Formula

$$C = (p + k) \bmod 26$$

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

a) Demonstrate the encryption process

plaintext	H	E	L	L	O
plaintext value (P)	7	4	11	11	14
key	K	E	Y	K	E
key value (K)	10	4	24	10	4
Ciphertext value (E)	17	8	$\frac{35}{9} \bmod 26$	21	18
Ciphertext	R	I	J	V	S

HELLO $\rightarrow$ RIJVS

b) Effect of repeating keyword

The keyword KEY is repeated

KEY $\rightarrow$ KEYKE

Each key letter is applied to the corresponding plaintext

plaintext	H	E	L	L	O
key	K	E	Y	K	E
cipher	R	I	J	V	S

* The keyword controls the amount of shifting for each character.

* The keyword KEY is repeated when the message is longer than the keyword. Each repeated key letter determines the shift used to encrypt the corresponding plaintext letter.

Result:

HELLO → RIJVS

2) Intercepted suspicious message - Substitution Cipher

Definition

A Substitution Cipher is a cryptographic technique in which letter of the original message (plaintext) is replaced by another letter according to a fixed substitution rule.

→ In this question, the ciphertext can be decoded using the Atbash substitution method.

What is Atbash Cipher?

Atbash Cipher is a Simple Substitution Cipher in which the Alphabet is written in reverse order.

∴

A → Z

B → Y

C → X

D → W

E → V

F → U

G → T

H → S

I → R

J → Q

K → P

L → O

M → N

The same rule is used in reverse for decryption.

Given Ciphertext

GSRH RH Z HVXIVG

We need to determine the original message using the

Substitution rule and the patterns in the Ciphertext.

Step 1 - Analyse the word patterns

The ciphertext contains four words:

GSRH | RH | Z | HVXIVG

Their lengths are

4, 2, 1, 6 letters

The pattern is: 4 - 2 - 1 - 6

A common English sentence with this structure is:

This is a secret

Step 2 - Decode the first word

Ciphertext: GSRH

using Albash:

G → T, S → H, R → I, H → S

∴ GSRH → THIS

Step 3: Decode the second word

R → I, H → S

∴ RH → IS

Step 4: Decode the third word

Z → A

Step 5: Decode the fourth word

HVXIVG

using the Albash rule:

H → S, V → E, X → C, I → R, V → E, G → T

∴ HVXIVG → SECRET

Step 6 - Combine the decoded

Ciphertext	plaintext
GSRH	THIS
RH	IS
Z	A
HVXIVG	SECRET

Final:

GSRH RH Z HVXIVG

THIS IS A SECRET